Memory Is All You Need

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Abstract

Transformers and state space models (e.g. Mamba) use sophisticated mechanisms to mix information across sequence positions. We argue that **a minimal mixing backbone and an associative memory is all that is necessary**. We augment language models with a Hebbian memory—a matrix of outer-product associations updated at each token—and systematically ablate the backbone. On The Stack (code), at 100M parameters: Mamba alone achieves 2.53 nats; adding Hebbian memory improves it to 1.82 nats; replacing Mamba's recurrence with a 4-token depthwise convolution gives 1.79 nats with no recurrence at all. The sequence model backbone contributes only local feature extraction; **the associative memory handles all long-range recall**.

1 Introduction

Transformers [Vaswani et al., 2017] and state space models like Mamba [Gu and Dao, 2024, Dao and Gu, 2024] use complex mechanisms—quadratic attention, selective gating, discretized recurrence—to capture long-range dependencies in sequences.

But what if associative memory, not the sequence model, is the primary mechanism for long-range reasoning? In this work, we decompose language modeling into two capabilities: first, **associative binding**—storing and retrieving associations across arbitrary distances—and second, **local feature extraction**—combining adjacent tokens into useful representations.

A Hebbian associative memory [Hebb, 1949]—a decaying outer-product matrix updated at each token—handles the first entirely. For the second, a depthwise convolution over just 4 tokens suffices. The resulting architecture contains no recurrence, no attention, and no gating beyond SiLU, yet matches SSM-based models at comparable parameter counts.

Our key finding is stark: replacing Mamba with a 4-token convolution matches performance (1.79 vs 1.82 nats) at a fraction of the complexity, and a model with *no* sequence mixing at all (pure linear projection + Hebbian) still converges to 3.18 nats.

The associative memory is not merely an augmentation—it is retrieval and computation in one mechanism. It *is* the model.

2 Method

The model (Figure 1) stacks L identical layers with pre-norm residual connections:

$$\mathbf{x}^{(l)} = \mathbf{x}^{(l-1)} + \text{HebbianMemory}(\text{Conv}(\text{RMSNorm}(\mathbf{x}^{(l-1)}))) \tag{1}$$

Each layer contains a local feature extractor (depthwise convolution) and a Hebbian associative memory. Token embeddings are tied with the output projection.

Parameter count For our 100M model (L=12, D=1024, E=2, d_{conv}=4), per layer:

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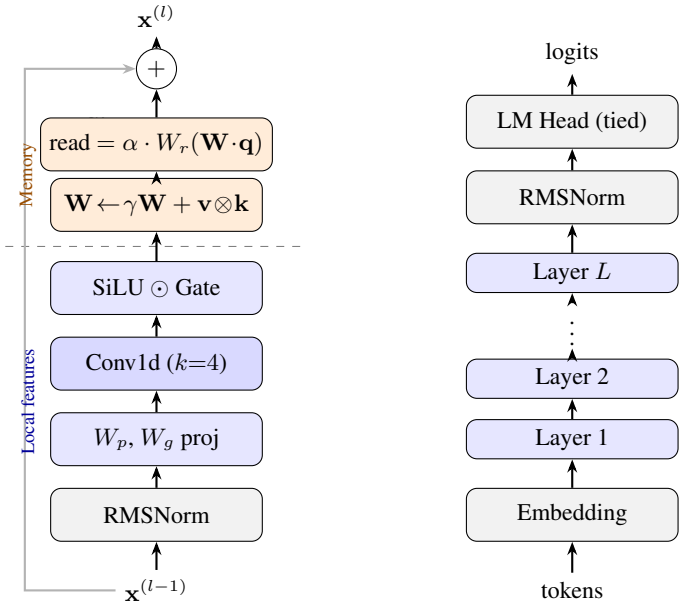

Figure 1: Architecture overview. **Left**: A single layer. The conv block (blue) extracts local features over 4 tokens. The Hebbian memory (orange) writes and reads associative bindings via a $D \times D$ outer-product matrix. **Right**: The full model stacks L identical layers with tied embeddings.

- Conv block: $3ED^2 + ED \cdot d_{\text{conv}} \approx 6.3$M (projections + conv kernel)
- Hebbian memory: $2D^2 + 2 \approx 2.1$M (write/read projections + decay/alpha)
- Total per layer: $\sim$8.4M; 12 layers: $\sim$100M

2.1 Hebbian Associative Memory

At each position t, the model maintains a memory matrix $\mathbf{W}_t \in \mathbb{R}^{D \times D}$ that accumulates outer-product associations between consecutive representations:

$$\mathbf{W}_t = \gamma \mathbf{W}_{t-1} + \mathbf{v}_t \otimes \mathbf{k}_{t-1} \tag{2}$$

$$\text{read}_t = \mathbf{W}_{t-1} \cdot \mathbf{q}_t \tag{3}$$

where $\mathbf{v}_t = W_v \mathbf{x}_t$ is the value projection, $\mathbf{k}_t = \mathbf{x}_t$ is the write key (the raw hidden state), $\mathbf{q}_t = \mathbf{x}_t$ is the read query, and $\gamma = \sigma(\delta)$ is a learned decay with δ initialized to 4.6 ($\gamma \approx 0.99$). Here $\mathbf{x}_t$ denotes the output of the conv block at position t. The layer output combines the conv output with the memory read:

$$\mathbf{y}_t = \mathbf{x}_t + \alpha \cdot W_r \cdot \text{read}_t \tag{4}$$

where $\alpha = \exp(\log \alpha_0)$ is a learned per-layer memory strength (initialized at 0.03) and W_r is a learned read projection. A residual connection from the layer input is added on top. Both α and γ are learned per layer.

For training, we use a chunkwise parallel form with $O(TC \cdot D + T \cdot D^2)$ time; at inference, the recurrent form (Equations 2–3) runs in $O(D^2)$ per step.

2.2 Convolution Backbone

Each layer's local feature extractor is a gated depthwise convolution:

$$\mathbf{x}' = \text{SiLU}(\text{Conv1d}(W_p \mathbf{x})) \odot \text{SiLU}(W_g \mathbf{x}) \tag{5}$$

$$\text{out} = W_o \mathbf{x}' \tag{6}$$

where $W_p, W_g \in \mathbb{R}^{D \times ED}$ project to an expanded dimension (E=2), Conv1d is a causal depthwise convolution with kernel size d_{conv}=4, and $W_o \in \mathbb{R}^{ED \times D}$ projects back. This block sees exactly d_{conv} tokens of local context. All long-range information flows through the Hebbian memory.

3 Experiments

All models are $\sim$100M parameters (12 layers, d_{model}=1024), trained on The Stack for 4000 steps with sequence length 2048, batch size 1, cosine schedule with peak 3×10^{-4} and 500-step warmup. We report average validation loss over steps 2000–4000 in nats (natural log cross-entropy). Validation is noisy at batch size 1, so we average across eval points.

3.1 Backbone Ablation

Our central experiment replaces the sequence modeling backbone while keeping the Hebbian memory identical. Table 1 shows results.

Table 1: Backbone ablation on The Stack ($\sim$100M params, 4000 steps). Mamba+Heb uses fixed α=0.04, γ=0.99. Conv+Heb uses learned per-layer α (init 0.03) and γ (init 0.99).

Model	Layers	Params	Avg Val Loss
Conv(4) + Heb	12	102.9M	1.79
Mamba + Heb	11	98.5M	1.82
Mamba (no Heb)	15	102.1M	2.53

Hebbian is the dominant mechanism. Removing Hebbian memory from Mamba degrades performance from 1.82 to 2.53 nats (+0.71). The memory contributes far more than the backbone.

Conv matches Mamba with no recurrence. Conv+Heb achieves 1.79 nats, matching Mamba+Heb at 1.82 nats. The convolution has zero recurrent state yet matches a full state space model.

Figure 2: Training curves on The Stack ($\sim$100M params, 4000 steps). Solid lines show smoothed training loss; markers show validation loss. Conv+Heb tracks Mamba+Heb throughout training, while Mamba without Hebbian memory lags by $\sim$0.7 nats.

3.2 Recurrent Inference

We evaluate both models using their recurrent (autoregressive) form on 4096-token windows from The Stack validation set, averaging over 8 windows. Table 2 shows per-segment losses.

Conv+Heb maintains a 0.54-nat advantage across all positions, confirming that the training-time advantage transfers to autoregressive generation. Both models show stable loss across sequence length—neither degrades at longer positions.

Table 2: Recurrent inference on The Stack (4096 tokens, 8 windows). Loss in nats by position segment.

Model	**0–1K**	**1–2K**	**2–3K**	**3–4K**	**Overall**
Mamba (no Heb)	1.98	1.87	2.05	1.99	1.97
Conv+Heb	1.46	1.35	1.47	1.44	**1.43**

3.3 Learned Memory Parameters

The Conv+Heb model learns per-layer memory strength α and decay γ. Figure 3 shows the learned values after 4000 steps.

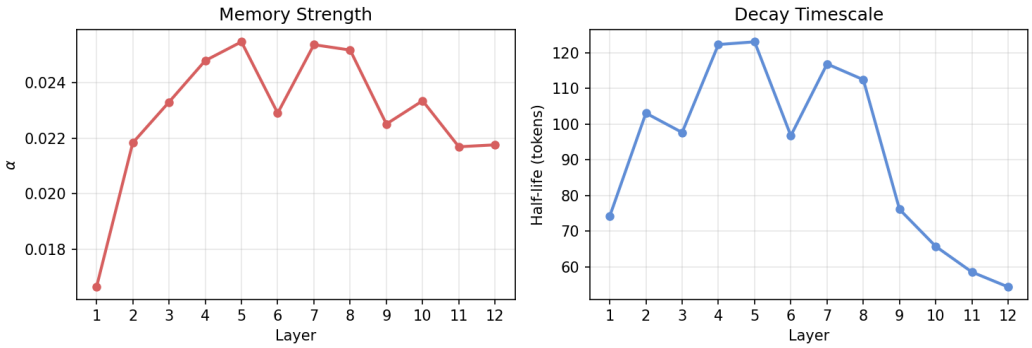

Figure 3: Learned per-layer parameters in the 100M Conv+Heb model. **Left**: Memory strength α peaks in middle layers ($\sim$0.025) and drops at boundaries ($\sim$0.017). **Right**: Decay half-life ($\log 2/\log(1/\gamma)$) shows a timescale hierarchy—early layers retain memory longer ($\sim$120 tokens), later layers decay faster ($\sim$55 tokens).

The α profile suggests middle layers perform the bulk of associative reasoning, while boundary layers focus on local features (early) and decoding (late). The decay hierarchy means early layers build broad context while later layers focus on recent associations.

4 Discussion

Hebbian memory as the dominant mechanism The $+0.71$ nat gap between Mamba and Mamba+Heb versus the negligible gap between Mamba+Heb and Conv+Heb points to a clear decomposition: the memory is doing the work. The outer-product matrix is content-addressable—retrieval is by similarity to the query, not by temporal position—making it fundamentally different from SSM recurrence, which compresses history into a fixed-size state vector.

This echoes complementary learning systems in neuroscience [McClelland et al., 1995]: neocortex for slow feature learning, hippocampus for fast associative binding. The Hebbian memory plays the hippocampal role.

Connection to fast weight programmers and linear attention Our memory is an instance of fast weight programming [Schmidhuber, 1992, Ba et al., 2016, Schlag et al., 2021]. Linear attention [Katharopoulos et al., 2020] maintains the same outer-product matrix without decay; adding learnable decay connects to RetNet [Sun et al., 2023] and GLA [Yang et al., 2024]. Our contribution is not the mechanism itself, but the finding that *it alone, paired with a trivial backbone, suffices for competitive language modeling*.

Limitations Our experiments are limited to code (The Stack) at 100M parameters and $\sim$8M tokens. Natural language remains untested. The $O(D^2)$ memory state per layer limits scaling to very large hidden dimensions.

5 Related Work

State space models S4 [Gu et al., 2022] introduced structured state spaces for sequence modeling. Mamba [Gu and Dao, 2024] added input-dependent selection, achieving strong performance on language modeling. Mamba-2/SSD [Dao and Gu, 2024] revealed the duality between SSMs and structured attention, enabling efficient multi-head implementations. Our work shows that for models augmented with associative memory, the SSM backbone can be replaced with a simpler alternative.

Associative memory and fast weights Our Hebbian memory follows the fast weight lineage [Schmidhuber, 1992, Ba et al., 2016, Schlag et al., 2021] and is closely related to linear attention [Katharopoulos et al., 2020], RetNet [Sun et al., 2023], GLA [Yang et al., 2024], and RWKV [Peng et al., 2023]. Hopfield networks [Hopfield, 1982, Ramsauer et al., 2021] provide an energy-based alternative. Our contribution is showing that these mechanisms make the surrounding architecture largely redundant.

6 Conclusion

A 4-token depthwise convolution paired with Hebbian outer-product memory beats a 100M-parameter Mamba on code language modeling—with no recurrence. Removing the memory costs 0.71 nats. Removing the SSM costs nothing. The backbone is not the model—the memory is.

The complexity of modern SSMs and attention may be solving a problem—long-range sequence modeling—that is better addressed by explicit associative storage. Future architecture work should focus on the memory mechanism, not the backbone.

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